QBEAST AI Circuit & Liquidity Flags: Technical and Functional Guide

Sep 27, 2026 · @Dheerendra Kumar Yadav

1. Overview

Four flags tell a backtest, paper or live system when a printed price was not really tradable, so it never books a fill that could not have happened. They are implemented in one file, `circuit_liquidity_flags.py` (version 2.4.1), and validated on 1,272 NSE/BSE stocks and 15 Nifty indices from 2000 to 2026.

#	Flag	Question it answers	Function
1	Price Band	What % move is this stock allowed today?	`compute_price_band_flag`
2	Upper / Lower Circuit	Did the stock close locked at its band today?	`compute_circuit_hit_flag`
3	MWCB	Did the whole market hit a circuit breaker today?	`compute_mwcb_flag`
4	Staircase	Is the stock in a run of daily circuits on thin trading?	`compute_staircase_flag`

On top of the four flags, a single gate column, `TRADE_BLOCKED`, combines staircase, chronic illiquidity and MWCB into one yes/no decision for the strategy.

Why it matters. Illiquid stocks can rise or fall by their full band every day on almost no volume. The printed close exists only as an unmatched order queue. A naive backtest buys at that price, shows large profits, and the live system can never reproduce them. INDOTHAI in December 2021 is the reference case: 15 consecutive +5% days, most of them with open = high = low = close.

Who this guide is for. Functional readers (strategy, risk) should read sections 2 to 7 for what each flag means and how to act on it. Technical readers (developers, data engineers) should also read sections 8 to 10 for the calls, inputs, outputs and parameters.

Files in the package (`E:\Downloads\QBEAST_circuit_flags_v2`, also in `QBEAST_circuit_flags_v2.4.1.zip`):

File	Purpose
`circuit_liquidity_flags.py`	The library: all 4 functions plus the gate, chronic flag and bar-by-bar tracker
`qbeast_flags_config.json`	The config: paths, workers and every flag parameter (loaded by `flags_config.py`)
`flags_io.py`	Shared CSV loaders and the complete EOD / intraday pipelines used by all scripts
`flags_for_one_day.py`	One stock, one day, each flag from its own call
`daily_backtest_loop.py`	Day-by-day causal loop over the whole universe, with a strategy hook
`run_four_flags.py`	25-worker run writing the 4 flag columns + `TRADE_BLOCKED` for every stock and index
`call_four_functions.py`	Shows each function called and what it returns
`verify_install.py`	End-to-end self-test on synthetic data; run it first on every new machine
`test_circuit_liquidity_flags.py`	27 automated tests
`README.md`, `QBEAST_AI_Circuit_Liquidity_Flags_SPEC_v2.md`, `CHANGELOG.md`	Call reference with every parameter, design specification, version history

2. Core rules

Every flag on day t is computed from bars up to and including day t, and describes the state at the close of day t. A decision made on day t fills at day t+1's open.

No forward bias. No row is ever re-marked after the fact, no end date is back-dated, and no window looks ahead. Two automated checks prove this on the real data for all 1,272 stocks:

- **No-repaint:** cutting the data at any day never changes a flag on or before that day.

- **Streaming parity:** feeding bars one at a time gives exactly the same flags as one call on the full history.

The only look at day t+1. The execution simulator may read the next bar's `open_at_upper` / `open_at_lower`, to decide whether a fill was physically possible at that open. The exchange does the same to a live order. The strategy itself never reads the next bar.

Trading rule while a stock is blocked (`TRADE_BLOCKED` not `NO`):

- no new buy
- no new sell
- no exit of an existing position
- mark-to-market continues, whether in profit or loss

The block lifts only causally. For a staircase, that means 2 consecutive normal days with real volume.

EOD vs intraday.

Flag	EOD (daily bars)	Intraday (1-minute or tick bars)
Price Band	per session, inferred or supplied	session band taken from the previous close
Circuit	once per session, at the close	per bar; a freeze = pinned at the band for 3+ bars; resets every session
MWCB	day-level flag from the index's high/low	halt and re-open auction windows by timestamp
Staircase	computed here	inherited from the previous close, plus an early block

Exchange preference. NSE is preferred; BSE is used when a stock trades only there. MCX and NCDEX commodity limits are out of scope.

3. Flag 1: Price Band

The price band is the maximum % a stock may move from the previous close today: 2%, 5%, 10% or 20%, or no band for F&O stocks. It is a rule, not an event; the only thing that happens to it is the exchange changing it.

Functional view

- The band sets the upper and lower circuit prices: upper = previous close × (1 + band), lower = previous close × (1 − band), rounded down to the tick.
- A narrowing band (20% → 10% → 5% → 2%) is an early warning that surveillance is tightening, ahead of a full freeze.
- Under the strictest GSM stage the exchange can forbid any rise while still allowing a 5% fall. That is an asymmetric band, upper 0 / lower 5, and is handled explicitly.

Technical view

`compute_price_band_flag(df)` is used when you have point-in-time band data.

	Columns
Input (required)	`symbol`, `date`, `upper_band_pct`, `lower_band_pct` (0 to 100; `inf` = no band; NaN = unknown)
Output (added)	`band_known`, `band_asymmetric`, `band_tightened`, `band_widened`

`band_tightened` compares today's band with the last known value only, so a missing day never hides a change.

Without band data (the case for Final_Data). Flag 2 infers the band causally, per stock and exchange. A day that closes locked at a move matching 2 / 5 / 10 / 20% (after tick rounding) sets the believed band. A later day that trades beyond that band resets it to unknown. The band believed for tomorrow is `band_est_next`; the readable value is the `PRICE_BAND` column.

PRICE_BAND value	Meaning
`2`, `5`, `10`, `20`	band in force for this session
`0/5`	asymmetric band from supplied data (upper / lower)
`NONE`	no band (supplied as `inf`, e.g. F&O stocks)
`UNKNOWN`	no band data and no locked day seen yet

Use cases

1. **Surveillance early warning.** A stock whose band goes 20 → 10 → 5 within a few weeks is moving into GSM/ASM. A strategy can reduce position size before any circuit prints.
2. **Setting realistic targets and stops.** A 5% band means a stop 8% below the price may be jumped in one gap; the stop fills at the lower circuit, or not at all if the stock locks.
3. **Inferred band on INDOTHAI.** The band read 10% until 17 Dec 2021 (it rose 10% that day). From 20 Dec every locked day moved ±5%, so the band switched to 5% without any external data.

4. Flag 2: Upper / Lower Circuit

A circuit hit means the stock closed locked at its band limit: at the upper limit there were only buyers, at the lower limit only sellers. On such a day a buy at the upper circuit (or a sell at the lower) usually does not fill.

Functional view

- **Upper circuit** (`UPPER`): closed at +band and at the day's high. Buying is effectively impossible; selling is easy.
- **Lower circuit** (`LOWER`): closed at −band and at the day's low. Selling is effectively impossible, so a holder is trapped.
- **Penny stocks** (`UPPER_1TICK` / `LOWER_1TICK`): when the band is under 2 ticks wide (a ₹1.05 stock with a 5% band = one ₹0.05 tick), every one-tick close at the high "hits" it. These are reported for information only and never feed the staircase.
- On its own a circuit day is a risk record, not a trading block. It feeds the staircase (Flag 4) and chronic illiquidity checks.

Technical view

`compute_circuit_hit_flag(df, mode="eod")` or `mode="intraday"`.

	Columns
Input (required)	`symbol` , `date` , `open` , `high` , `low` , `close` , `volume`
Input (optional)	`prev_close` (else derived), `upper_band_pct` / `lower_band_pct` (else inferred), `exchange` (a bar right after an exchange switch is marked unknown), `_gap_filled` (forward-filled rows are unknown)

	Columns
EOD output	`circuit_locked_dir` (upper / lower), `band_hit_dir`, `frozen` (O = H = L = C), `single_price`, `tick_degenerate`, `band_upper_used`, `band_est_next`, `open_at_upper` / `open_at_lower`, `data_unknown`
Intraday output	`pinned`, `pinned_run_len` (resets each session), `intraday_freeze` (pinned 3+ bars), `freeze_dir`, `locked_since_open_dir`

How a hit is decided (EOD).

1. Move = close / previous close − 1.
2. Tolerance = one tick (₹0.05) / previous close. NSE rounds limits down to the tick, so a 5% band on ₹1.70 allows only +2.94%. For back-adjusted prices off the tick grid, the tolerance is capped at 0.5%.
3. A hit = the move is within tolerance of the band **and** the stock is locked: close = high for upper, close = low for lower (`close_locked`, the default). The stricter `lock_mode="strict"` requires O = H = L = C.

Intraday guards. A bar counts as pinned only if the session has not already traded beyond that band, and the band must be known from the EOD history (no guessing).

Use cases

1. **Fill feasibility in the simulator.** If the next day opens locked at the upper circuit (`open_at_upper` on day t+1), a buy order placed on day t does not fill. The order stays pending.
2. **Trapped-position risk.** INDOTHAI 11 to 14 Jan 2022: four lower circuits in a row, three with O = H = L = C. A long position could not be exited at any price.
3. **Tick rounding caught correctly.** GTLINFRA frozen days at +2.6%, +2.9% and +3.4% are all 5% bands rounded down to the ₹0.05 tick; a plain % comparison would miss every one of them.
4. **Partial-day circuits (intraday).** 63MOONS froze at its band in 229 sessions between 2022 and 2026, including sessions where it later unlocked. An EOD-only check cannot see those.

5. Flag 3: MWCB (market-wide circuit breaker)

MWCB halts every stock and F&O contract at once when NIFTY 50 or SENSEX moves 10%, 15% or 20% from its reference close. It is a rare, market-wide event; in the data it fired on 10 days between 2000 and 2026.

Functional view

- It is one state for the whole market: every stock inherits the same MWCB value on a given date.
- The trigger is the index's intraday extreme, not its close. On 13 Mar 2020 Nifty fell 10.8% at the low (a 45-minute halt) and then closed +3.8%; a close-only test misses it.
- After each halt, trading re-opens through a 15-minute pre-open call auction, not normal continuous trading. The re-open price comes from a thin auction.

Halt rules (current SEBI regime; configurable):

Move	Before 13:00	13:00 to 14:00	14:00 to 14:30	After 14:30
10%	45 min	15 min	15 min	no halt
15%	1 h 45 min	45 min	rest of day	rest of day
20%	rest of day	rest of day	rest of day	rest of day

Technical view

`compute_mwcb_flag(index_df, mode="eod")` or `mode="intraday"`, then `aggregate_mwcb_by_date()` before joining onto stocks. Joining two indices on date without aggregating would duplicate every stock row.

	Columns
Input (required)	`index_name`, `date`, `close`; `high` and `low` strongly recommended
EOD output	`mwcb_level` (0 / 10 / 15 / 20), `mwcb_dir` (upper / lower), `mwcb_triggered`, `extreme_up_pct`, `extreme_dn_pct`
Intraday output	`mwcb_halt_active`, `mwcb_reopen_auction`, `mwcb_halt_rest_of_day`, `mwcb_trigger_level`
Readable column	`MWCB`: `LOWER_10`, `LOWER_15`, `UPPER_20`, or blank (EOD); `HALT_10` / `AUCTION_10` or `NA` (intraday)

Threshold basis. The limit in index points = % × reference close. Before 1 Jul 2013 the reference is the previous quarter's last close; after it, the previous day's close. The history supports this split: 17 Oct 2007 and the 20% level on 18 May 2009 appear only on the quarterly basis, and 13 Mar 2020 only on the daily basis. The exact cut-over date still needs checking against the SEBI circular.

Intraday caution. An EOD MWCB value is never stamped on intraday bars of the same day: it is only known at that day's close. Intraday MWCB needs intraday index data; without it the column shows `NA`.

Use cases

1. **Blocking trades on a crash day.** RELIANCE on 13 Mar 2020 shows `MWCB = LOWER_10` and `TRADE_BLOCKED = MWCB_DAY`. No new orders are placed from that day's close.
2. **Intraday fill deferral.** A buy signal at 09:25 on a halt day cannot fill until the re-open auction ends. The simulator should use the first bar after the auction, not the next chronological minute.
3. **Stress testing.** With only 10 events in 26 years, most backtest windows contain no MWCB at all. Stop-loss and margin logic should be tested with a synthetic halt and re-open gap.

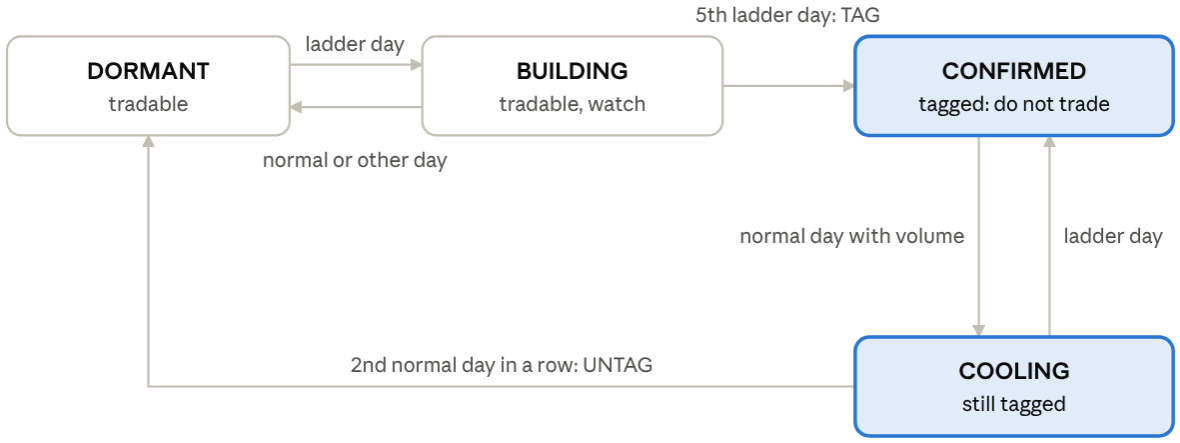
Detected days (NIFTY 50): 17 May 2004 (15%), 22 May 2006, 17 Oct 2007, 21 Jan 2008, 22 Jan 2008, 24 Oct 2008, 18 May 2009 (20% up), 5 Oct 2012 (15%), 13 Mar 2020, 23 Mar 2020 (all 10% down unless noted). 21 Jan 2008 and 24 Oct 2008 still need confirmation.

6. Flag 4: Staircase

A staircase is a run of daily circuits in the same direction, day after day: the "circuit ladder" of illiquid stocks. Once identified, the stock is not traded in backtest, paper or live until it returns to normal trading.

A stock is tagged on its 5th ladder day and untagged after 2 normal days

Unknown days change nothing; a direction flip while tagged keeps the tag



staircase state machine · 4 states

A stock is tagged at the close of its 5th consecutive same-direction ladder day, and stays tagged until 2 consecutive normal days with real volume.

Functional view

Each day is put into one of four classes:

Day class	Condition
LADDER	closed locked at the band (Flag 2), not a 1-tick penny hit, not upside-locked
NORMAL	not locked, not a single-price day, volume at least 0.5 × the stock's recent typical volume
UNKNOWN	gap-filled row, or the first bar after an exchange switch; changes nothing
OTHER	anything else (a thin-volume day, a single-price day)

Rules.

- The tag starts on the 5th ladder day itself. Earlier days are never marked afterwards, because that would use future information.
- While tagged, an OTHER day (thin volume) resets the normal-day count, so a dead stock stays blocked.
- A direction flip while tagged (climbing into lower circuits) keeps the tag and starts a new episode. CLIMBING and FALLING staircases are blocked identically: no buy and no sell.
- The volume baseline is the median of the last 20 clean trading days. It restarts when the exchange changes (NSE and BSE volumes differ in scale).

Technical view

`compute_staircase_flag(circuit_output)` for EOD; `compute_staircase_flag(intraday_circuit_output, mode="intraday", eod_state_df=...)` for intraday.

	Columns
Input	the output of `compute_circuit_hit_flag` (EOD or intraday)
EOD output	`staircase` (True = tagged), `staircase_state`, `staircase_dir` (climbing / falling), `ladder_run_len`, `normal_day_count`, `staircase_event_id`, `day_class`, `vol_baseline`, `staircase_early_block_dir`
Intraday output	`staircase_intraday`, `eod_staircase_prev`, `early_block`
Readable column	`STAIRCASE`: `CLIMBING` / `FALLING` (tagged, do not trade), `BUILDING` (watch), blank; intraday `EARLY_BLOCK`

Intraday. Each bar inherits the tag as of the previous close. Early block: if yesterday closed one ladder day short of 5, and every bar today so far has been pinned at the band in that direction, the stock is blocked for the rest of today. Untagging never happens intraday.

Use cases

1. **INDOTHAI, Dec 2021 to Mar 2022.** BUILDING from 17 Dec. CLIMBING from 23 Dec (5th upper circuit). It flipped to FALLING on 11 Jan 2022 and stayed tagged until 17 Mar 2022, because normal days with volume did not return. Nothing was traded in that period.
2. **The famous multibaggers.** MIDWEST (₹53 to ₹3,403 with a 103-day run), GHVINFRA (201 circuits in a row), SABTNL, KOTIC and DIACABS are all tagged for most of their runs; a backtest cannot book those gains.
3. **Watch list.** On the last day of the data, 7 stocks were BUILDING (for example IDEAFORGE, HFCL, E2E) and 3 were tagged (RELINFRA, STLTECH, A1L). BUILDING stocks are still tradable, but one or two more circuits would block them.
4. **Holding a stock that becomes tagged.** The position is kept and marked to market daily. No exit is attempted until the tag lifts, which mirrors reality: a stock in a falling ladder cannot be sold.

7. The trading gate: TRADE_BLOCKED and chronic illiquidity

`TRADE_BLOCKED` is the one column a strategy needs to read: `NO` means tradable at the next open; anything else means no buy, no sell and no exit.

Value	Meaning	Source
`NO`	tradable	–
`STAIRCASE`	tagged staircase (CLIMBING or FALLING)	Flag 4
`CHRONIC`	chronic illiquidity over the last 60 sessions	chronic flag
`MWCB_DAY`	market-wide circuit breaker day (EOD)	Flag 3
`MWCB_HALT`	inside a halt or re-open auction window (intraday)	Flag 3

Several reasons are joined with `+`, for example `STAIRCASE+CHRONIC`. `BUILDING` is not a block. Index files show `NA`, because indices are not traded.

Chronic illiquidity (`compute_chronic_illiquidity_flag`) looks at the last 60 sessions and tags the stock when any of these is true:

Measure	Tag when at least	Untag when below
Single-price days (O = H = L = C, not at a band)	20%	10%

Measure	Tag when at least	Untag when below
No-trade days (zero volume, gap-filled, or a trading day missing from the file)	20%	10%
Circuit-locked days	30%	15%

The lower untag level stops the flag switching on and off every day. It catches dead stocks that never form a clean staircase, such as INDOTHAI in 2012 to 2016 with its many single-price days.

Fill feasibility (for the execution simulator only). `buy_fill_feasible_at_open` and `sell_fill_feasible_at_open` describe a bar's own open. The simulator reads them from the fill bar (t+1): a buy cannot fill at an open locked at the upper band, a sell cannot fill at one locked at the lower band, and nothing fills inside an MWCB halt.

Use cases

1. **Daily gate in a backtest.** After each close, skip every stock whose `TRADE_BLOCKED` is not `NO`; keep marking existing positions to market.
2. **Universe size.** Across 4.14 million stock-days, 151,551 (3.66%) are blocked: 146,079 by staircase or chronic and 5,472 by MWCB days alone.
3. **Frequent-circuit stocks.** 63MOONS is blocked as `CHRONIC` on about 32% of its intraday bars because it was circuit-locked on more than 30% of recent sessions in 2022 to 2024. This is intended: a stock that locks that often is a circuit trap even when its volume is high.

8. How to call the flags

There are three equivalent ways to call the flags, and they give identical answers: one stock on one day, a day-by-day loop, or one batch call read row by row. Commands below run in PowerShell from `E:\Downloads\QBEAST_circuit_flags_v2`; code blocks marked Python go inside a `.py` file, never into PowerShell.

A. One stock, one day (after the close)

```
python flags_for_one_day.py --symbol INDOTHAI                        # latest
day (live use)
python flags_for_one_day.py --symbol INDOTHAI --date 2021-12-24     # a past
day
python flags_for_one_day.py --symbol INDOTHAI --from 2021-12-15 --to 2022-01-
20 --verify   # daily loop
```

Each flag has its own call, given history up to and including the day being checked (Python):

```
from flags_for_one_day import get_price_band, get_circuit, get_mwcb,
get_staircase, get_trade_blocked

s = stock[stock.date <= day]    # the stock's rows up to and including today
n = nifty[nifty.date <= day]    # NIFTY 50 rows up to and including today

get_price_band(s)          # FLAG 1 -> "5" / "10" / "20" / "2" / "UNKNOWN"
get_circuit(s)             # FLAG 2 -> "UPPER" / "LOWER" / ""
get_mwcb(n)                # FLAG 3 -> "LOWER_10" / "UPPER_20" / ""
get_staircase(s)           # FLAG 4 -> "CLIMBING" / "FALLING" / "BUILDING" /
""
get_trade_blocked(s, n)    # gate   -> "NO" or "STAIRCASE" / "CHRONIC" /
"MWCB_DAY"
```

Example output for INDOTHAI on 24 Dec 2021: `PRICE_BAND 5 · CIRCUIT UPPER · MWCB – · STAIRCASE CLIMBING · TRADE_BLOCKED STAIRCASE`.

B. Day-by-day loop over the whole universe (backtest, paper, live)

```
python daily_backtest_loop.py --data "E:\Downloads\Final_Data" --workers 25 -
-verify
```

One `CircuitLiquidityTracker` per stock is fed one bar per day, so the work per day stays constant. Put the strategy in `strategy_hook()`; when `blocked` is True, do not open, add to or exit that position. At the end the tracker state is saved to `trackers_state.pkl`; reload it each morning in live trading. The core of the loop (Python):

```
tracker = C.CircuitLiquidityTracker("INDOTHAI")
for bar in bars:                      # one daily bar at a time, in date
order
    f = tracker.update_eod(bar)       # flags as of today's close
    blocked = f["staircase"] or f["chronic_illiquid"] or mwcb_today
```

C. One batch call, read row t on day t

```
python run_four_flags.py --data "E:\Downloads\Final_Data" --workers 25 --
intraday --check-every 1
```

This writes the 4 flag columns plus `TRADE_BLOCKED` for every stock (`eod_stocks\<SYMBOL>.csv`), index and intraday file into `Final_Data\_four_flags_out`. It is allowed because row t never uses a later bar, and the self-check re-proves that on every run.

The four functions directly (Python)

```
import circuit_liquidity_flags as C
flag1 = C.compute_price_band_flag(band_table)            # only with real
band data
flag2 = C.compute_circuit_hit_flag(stock, mode="eod")
flag3 = C.compute_mwcb_flag(nifty.assign(index_name="NIFTY50"))
flag4 = C.compute_staircase_flag(flag2)                  # needs Flag 2's
output
```

Method	Speed	Best for
A. one stock, one day	~0.1 s per call	a single stock or a watchlist, checking a day
B. daily loop with trackers	~30 µs per stock per day	full-universe backtests, paper and live
C. batch, read row t	1,272 stocks in ~110 s on 25 workers	research, universe statistics

9. Validation on Final_Data

Version 2.3.0 passed every check on the full data (run on 27 Sep 2026, 25 workers, 109 seconds).

Check	Result
Automated tests	24 / 24 passed
EOD stocks processed	1,272 / 1,272, 4,138,644 stock-days (2000 to 2026)
No-repaint check on all 5 columns	passed for all 1,272 stocks

Check	Result
Bar-by-bar tracker = batch	passed for all 1,272 stocks
Daily loop = batch (`daily_backtest_loop.py --verify`)	0 mismatches on the tested sample
Indices	15 Nifty index files
Intraday	10 files; 7 large/mid caps show no false freezes

Flag counts across all stock-days.

Column	Values
PRICE_BAND	UNKNOWN 3,810,623 (92%, no band data supplied) · 5%: 210,484 · 10%: 59,305 · 20%: 42,253 · 2%: 15,979
CIRCUIT	UPPER 42,603 · LOWER 23,421 · 1-tick penny hits 7,214
MWCB	5,707 stock-days on the 10 market-wide breaker days
STAIRCASE	CLIMBING 16,041 · FALLING 10,155 · BUILDING 47,855
TRADE_BLOCKED	151,551 (3.66%): CHRONIC 119,705 · STAIRCASE+CHRONIC 21,469 · MWCB_DAY 5,472 · STAIRCASE 4,670 · others 235

Staircase episodes: 1,416 in 347 stocks; half last 9 days or less, 90% under 40 days. Starts peak in 2020 (196) and 2021 (156), the small-cap boom years. RELIANCE is never tagged.

Bugs found and fixed during validation: 13, including long ladders cut short by a starved volume baseline, a missed MWCB on 13 Mar 2020, false circuits in back-adjusted history, false intraday freezes on large caps, and pandas swapping day and month on ISO dates.

10. Parameters, limits and open items

The defaults below were tuned on the full-universe runs; every one can be changed through a keyword argument without touching the code.

Parameter	Default	Flag	Effect
`start_confirm_days` (S)	5	Staircase	consecutive ladder days needed to tag
`end_confirm_days` (E)	2	Staircase	consecutive normal days needed to untag
`normal_vol_frac`	0.5	Staircase	a normal day needs this share of typical volume
`volume_lookback`	20	Staircase	clean days in the volume baseline
`vol_frac`	None	Staircase	optional thin-volume requirement for a ladder day
`lock_mode`	close_locked	Circuit	`strict` requires O = H = L = C
`tick_size` / `min_band_ticks`	₹0.05 / 2	Circuit	tolerance and the penny-stock guard
`intraday_min_pinned_bars`	3	Circuit (intraday)	bars pinned at the band for a freeze
`basis` / `quarterly_basis_before`	auto / 2013-07-01	MWCB	how the 10/15/20% limits are measured
`block_on_mwcb_day`	True	Gate	block the day after an MWCB day as well
chronic window / thresholds	60 sessions; 20% / 20% / 30%	Chronic	see section 7

Known limits.

- **No point-in-time band data** in Final_Data: bands are inferred from price action, so liquid and F&O stocks mostly show `UNKNOWN`.
- **F&O stocks** can show a false circuit when they touch their flexible 10% operating band (example: HFCL on the last day).
- **Back-adjusted prices** make circuit detection before about 2005 approximate.
- **SENSEX** is not in the data, so NIFTY 50 alone drives MWCB.
- **Intraday** is validated on 10 stocks only (two with real circuits: 63MOONS, AARTIPHARM).

- **Live use** needs the tracker state saved at the end of each day and reloaded the next morning.

Open items.

- [ ] Validate intraday on illiquid small-cap files (to be added to `Final_Clean_Stocks_INTRADAY_Data`)
- [ ] Confirm the SEBI date when MWCB limits switched from the quarterly to the daily reference close
- [ ] Confirm whether halts occurred on 21 Jan 2008 and 24 Oct 2008
- [ ] Optional: build point-in-time F&O membership from the F&O bhavcopy folders and pass it as "no band"